Gradient Descent: Theory and Implementation

A Comprehensive Guide

Exploring the mathematics and practical applications of gradient descent algorithms

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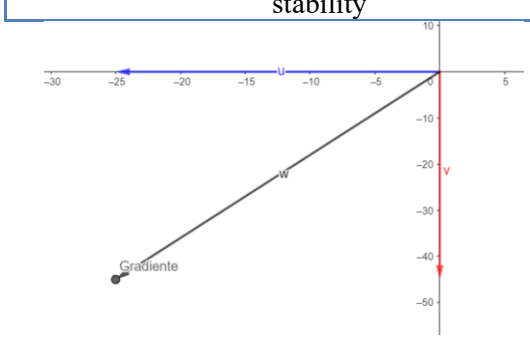
Introduction to Gradient Descent

Gradient descent is an optimization algorithm used to minimize a function by iteratively moving in the direction of the steepest descent, as defined by the negative gradient. This method is widely used in machine learning for training models, especially in linear regression and neural networks.

Comparison of Gradient Descent Types

Type	Description	Advantages	Disadvantages
Batch	Uses all training data for each update	Stable convergence	Computationally expensive
Stochastic	Uses single data point per update	Fast convergence	Noisy updates
Mini-batch	Uses small batches	Balances speed and stability	Requires tuning of batch size

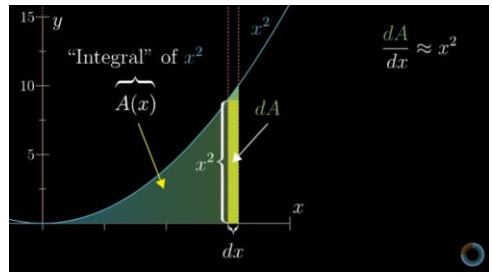


Gradient descent in a 2D cost function

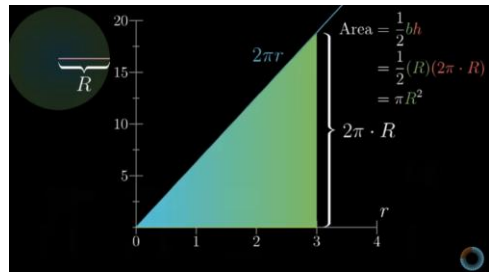
Mathematical Foundations

$$\theta_{t+1} = \theta_t - \alpha \nabla J(\theta_t)$$

Gradient descent update rule

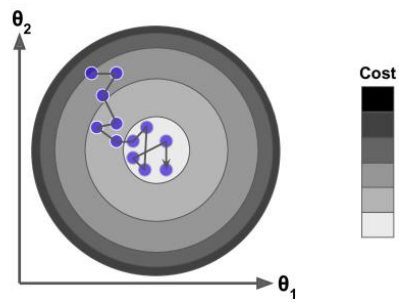


Derivative concept in gradient descent

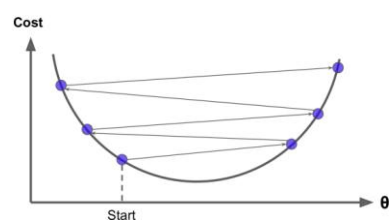


Area under the curve for integral approximation

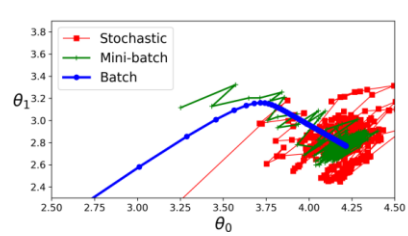
Types of Gradient Descent



Stochastic Gradient Descent visualization

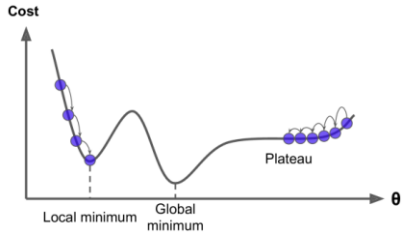


Learning rate impact on convergence

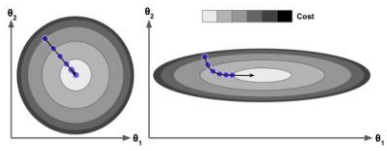


Comparison of different gradient descent methods

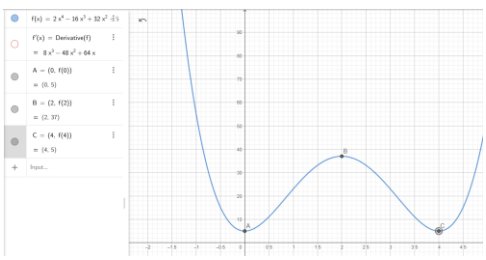
Common Pitfalls and Solutions



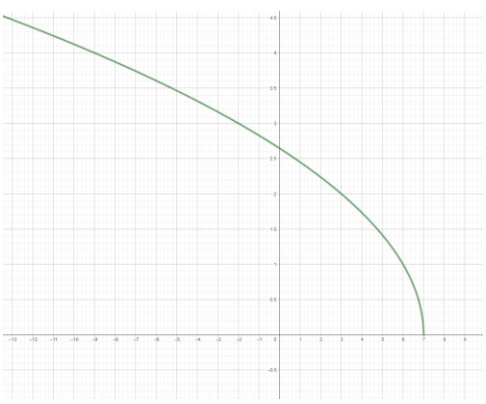
Gradient descent pitfalls like local minima



2D gradient descent with multiple features



First derivative test for minima



Domain of the function in gradient descent

Gradient descent is a fundamental optimization technique in machine learning. By iteratively moving in the direction of the negative gradient, it efficiently minimizes the cost function. However, careful tuning of the learning rate and understanding the landscape of the cost function are crucial for successful implementation.